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GraalLSP

GraalLSP: A Zero-overhead Polyglot Language Server Implementation for the GraalVM

GraalLSP: Eine Overhead-freie Language Server Implementierung für die GraalVM

by

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Abstract

The english abstract.

Zusammenfassung

Die Zusammenfassung auf deutsch.

Acknowledgments

I owe everything to my cat.

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LSP Language Server Protocol
IDE Integrated Development Environment

1. Introduction

- language development is challenging task
- what makes a good language: amongst performance, std lib, ecosystem and community, tools (for language developers and users) are important
- who provides the tools? language developers vs. language users
- normally tools are developed for a specific language and cannot be reused or need lots of adaptation
- In the Truffle framework, tooling (instrumentation) is part of the framework and therefore tools are implemented language-agnostic once
- Truffle has already some language-agnostic tools implemented (debugger, profiler, ...) and new ones can be easiliy built
- Truffle tooling support does not interfere with implementing high-performance languages
- Compared to a full IDE, some tools are missing
- Implementing LSP can provide further IDE-like features
- We propose an approach to build a generic LSP server on top of Truffle with:
  - minimal to zero language runtime performance overhead
  - minimal API changes to Truffle
  - minimal impl and maintainance overhead for language developers
- Benefits for the language developers: get IDE-like tools with minimal effort
- Benefits for language users: can choose from wide range of LSP clients + consistent dev-tools experience for different Truffle languages
- As proof of concept, we have implemented GraalLS - A Zero-overhead Polyglot Language Server Implementation for the GraalVM
- ?: focus on dynamically type languages? Or is this focus already included in Truffle (most impls are dynamically type langs)?

1.1. Contributions

- Contribution in detail:
  - An approach to build a generic LSP server on top of Truffle
  - An implementation of said approach: GraalLSP
  - Proposals to extend the Truffle Framework with new interfaces needed to support more LSP features
  - An experience report how well existing Graal/Truffle-implemented languages work with GraalLSP (with and without implementing the said interface proposals)

1.2. Structur of this Thesis

2. Context

2.1. ? Programming Language Concepts

- stuff needed to understand Truffle instrumentation + Approach (using SL or Python as example):
  - AST (+mapping to source section)
  - AST Interpreter
  - frames
  - scopes

2.2. Truffle and GraalVM

- Truffle language implementation framework
  - Truffle instrumentation
  - Loading languages (dynamic classpath detection)
- GraalVM - high-performance polyglot VM

2.3. Language Server Protocol

- definition
- client + server
- JSON-RPC
- graphic of exemplary LSP communication
- lsp4j - Java binding for LSP
  - version
  - map to Java types
  - cancelable requests via CompletableFuture
  - JSON-RPC wrap
- LSP client: VSCode example

3. Approach

- language implementaion + tool development = challenging task
- we propose: implement LSP for Truffle
- typically one specific language server impl for one language
- Our approach: a language-agnostic language server impl for multiple languages created by the language implementation framework Truffle

3.1. Differences to implementing a language-specific language server

- derive language from MIME-type
- completion trigger characters - lsp needs a fixed set at start-up -> provide all chars of all languages
- signatureHelp trigger characters - same as before
- lsp clients need set of supported languages already at extension/adapter definition - no dynamic look-up

3.2. Which data needs LSP server? <- move to context?

- LSP feature overview
- URIs to identify a source file and Locations
- data needed for: diagnostics, completion, goto-definition, ...
- on didOpen: content is sent once, then incremental updates -> store state

3.3. Use of existing Truffle-APIs

- what is already there?
- parsing sources
- AST
- scopes (global/local)
- frames and frame descriptors
- instrumentation
- -> functionality to support impl of debugger

3.4. What is missing?

3.4.1. Static information

- AST design focused on runtime, lacking many parse-time infos
  - mark declarations
  - mark variable read/write locations
  - ...
  - in general: semanitc relations between nodes
- most lang impls: no syntax-error tolerant parser

3.4.2. Proposed Truffle API changes

- language API
- Tags

3.5. How to deal with dynamically typed languages? -> Using runtime information

- Observe execution via instrumentation
- Side-effects -> Truffle-sandboxing
- Evaluation strategies:
  - Literals
  - Coverage data
    - storing frames
    - source location format
    - migrating coverage data on source edit
  - Run-to-section
  - Globals

3.6. Mapping LSP language server API to Truffle API

- diagnostics: parsing: throws exception on syntax error -> map to to source range -> send diagnostic, fail-on-first-error

3.7. Architecture

- graphic showing launcher, lsp server, truffle api communication
- comparison to other language server impls (built on top of language)

4. Implementation

4.1. The lsp4j library

4.2. Architecture

4.2.1. Truffle-API + Graal-SDK

4.2.2. Launcher, Instrument, Server

4.2.3. Thread with entered Context

4.3. What a TruffleLanguage needs to implement

The onInputValue method can be used in combination with source section filters to intercept very fine-grained execution events such as intermediate values used by language expressions. The intermediate values that are accessible to the instrumentation framework greatly depend on the instrumentation support provided by each language.

4.3.1. Expression- and Statement-Tag

4.3.2. findSourceLocation()

4.3.3. findTopScopes()

4.3.4. findLocalScopes()

4.3.5. findMetaObject()

4.3.6. getCompletionTriggerCharacters()

4.3.7. boxPrimitive()

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4.3.9. GetSignature custom message

4.3.10. ReadLocalVariableTag + WriteLocalVariableTag

4.4. LSP features implementation - TextDocumentService

4.5. LSP features implementation - WorkspaceService

4.6. LSP client config

4.6.1. VSCode extension exmaple + Limitations

5. Experience report

5.1. Between IDE and Language Workbench

Language Workbench focus/aim: Efficient language (DSL) development, performance is not main focus vs. Graal/Truffle aim: Efficient execution

Assumption: Language Workbenches use static information only, our Truffle-based solution uses dynamic type information

5.2. Completion response time

5.3. Limitations

own chapter? or in cha:implementation?

5.3.1. VSCode: Need to define supported languages in extension

5.3.2. VSCode: Syntax highlighting defined at client-side

5.3.3. LSP: No incremental sending of (completion) results

5.3.4. Server: Only TCP (connecting to already running server), no std-in/-out impl

5.3.5. Truffle: Only first syntax error is reported. Lang parsers fail on first error

6. Related Work

6.1. Graal

[5]

6.2. Squimera

[4] [3]

6.3. Language Workbenches

[1]

6.4. Run-time type collection

[2]
CTags, SemanticDB

7. Conclusion and Future Work

Bibliography

[1] Sebastian Erdweg, Tijs Van Der Storm, Markus Völter, Meinte Boersma, Remi Bosman, William R Cook, Albert Gerritsen, Angelo Hulshout, Steven Kelly, Alex Loh, et al. "The State of the Art in Language Workbenches". In: *International Conference on Software Language Engineering*. Springer. Springer International Publishing, 2013, pp. 197–217. DOI: 10.1007/978-3-319-02654-1_11.

[2] Michael Haupt, Michael Perscheid, and Robert Hirschfeld. "Type harvesting: a practical approach to obtaining typing information in dynamic programming languages". In: *Proceedings of the 2011 ACM Symposium on Applied Computing*. ACM. 2011, pp. 1282–1289. DOI: 10.1145/1982185.1982464.

[3] Fabio Niephaus, Tim Felgentreff, Tobias Pape, and Robert Hirschfeld. "Squeak Makes a Good Python Debugger: Bringing Other Programming Languages Into Smalltalk's Tools". In: *Companion to the first International Conference on the Art, Science and Engineering of Programming*. ACM. 2017, p. 27. DOI: 10.1145/3079368.3079402.

[4] Fabio Niephaus, Tim Felgentreff, Tobias Pape, Robert Hirschfeld, and Marcel Taeumel. "Live Multi-language Development and Runtime Environments". In: *arXiv preprint arXiv:1803.10200* 2.3 (Mar. 2018). DOI: 10.22152/programming-journal.org/2018/2/8.

[5] Chris Seaton, Michael L Van De Vanter, and Michael Haupt. "Debugging at Full Speed". In: *Proceedings of the Workshop on Dynamic Languages and Applications*. ACM. ACM Press, 2014, pp. 1–13. DOI: 10.1145/2617548.2617550.

Appendix A.

First Unimportant stuff.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Eigenständigkeitserklärung

Hiermit versichere ich, dass ich die vorliegende Arbeit selbständig verfasst sowie keine anderen Quellen und Hilfsmittel als die angegebenen benutzt habe.

Berlin, den 18. Oktober 2018

Daniel Stolpe

Todo list

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